

SADHWIKA R

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EDUCATION

University of Cincinnati, Ohio, USA | Master of Engineering in Computer Science | GPA: 3.75/4.0

May 2025

Jawaharlal Nehru Technological University, India | B. Tech in Computer Science | GPA: 3.5/4.0

May 2023

EXPERIENCE

BitByte Solutions LLC, Software Engineer, Philadelphia, USA

December 2024 - Present

- Engineered a transaction tracking module using Java (Spring Boot) and PostgreSQL, allowing users to view and search financial records in real-time, reducing manual reconciliation work by 25%.
- Built an interactive frontend using React.js, enabling users to generate reports and dashboards for cash flow analysis.
- Created RESTful APIs to support transaction workflows and optimized query execution paths, reducing latency from 2s to 700ms and improving backend responsiveness.
- Deployed AWS services including EC2 for backend hosting, RDS (PostgreSQL) for structured financial data storage, and S3 for report file management, supporting high availability and scalability.
- Collaborated in setting up CI/CD pipelines with GitHub Actions to resolve a recurring deployment bug that blocked releases, reducing release effort by 40% and restoring on-time delivery.

University of Cincinnati, Full Stack Developer-Graduate Student Assistant, Cincinnati, USA

September 2023 - December 2024

- Developed a role-based student portal using Next.js, Prisma, and PostgreSQL, enabling students, teachers, and admins to manage dashboards with secure access to subjects, classes, exams, and announcements.
- Implemented RESTful APIs to handle CRUD operations across user roles, ensuring reliable data exchange between frontend and backend, and maintaining data integrity across modular service layers.
- Integrated Clerk and optionally explored AWS Cognito for secure authentication and role-based authorization, ensuring protected access to sensitive user data across dynamic routes.
- Improved accessibility and engagement for students with cognitive learning needs by redesigning portal UI based on direct feedback and leveraging AWS PartyRock to deliver personalized academic support through adaptive content delivery.
- Conducted hands-on training sessions for over 50 students on Git version control, full-stack integration, and basic cloud deployment practices, mentoring peers on real-world software development workflows.

Forge Alumnus, Software Developer Intern, Telangana, India

March 2022 - April 2023

- Contributed to the development of a digital healthcare payment system using React.js, Node.js, and Spring Boot, improving transaction flow and reducing payment processing time by 30%.
- Enhanced designing and building RESTful APIs to manage patient payment data, incorporating basic access control to support secure data handling.
- Implemented MongoDB components and optimized queries to retrieve payment data, which improved data access speed and supported smoother transaction processing for high-volume records.
- Adopted autoscaling on AWS EC2 to handle fluctuating demand, reducing latency by 35% during high-traffic periods and maintaining system stability under peak load.
- Managed application deployment on AWS EC2 and S3 and employed New Relic to monitor key performance indicators including latency, system load, and error rates.

SKILLS

Programming & Frameworks: Java, Python, JavaScript, C++, SQL, Spring Boot, React.js, Node.js, Flask, HTML5, CSS3

Database & Development Tools: PostgreSQL, MySQL, MongoDB, Git, Jira, RESTful APIs, Agile (Scrum), SDLC

Cloud & DevOps: AWS, Azure, Docker, GitHub Actions, CI/CD, Linux

Certifications: [AWS Certified Solutions Architect – Associate](#), [Microsoft Certified: Azure Fundamentals](#)

PROJECTS

Uber Clone App:

- Built a full-featured Uber Clone using the MERN stack (MongoDB, Express, React, Node.js) with user and captain dashboards, Google Maps integration, ride booking logic, and real-time location features.
- Implemented secure user authentication, ride distance/time estimation with Google Maps API, and dynamic UI screens, delivering a scalable, production-grade ride-booking experience.

Potato Crop Disease Prediction:

- Developed an end-to-end deep learning pipeline using TensorFlow and CNNs to classify potato plant diseases from leaf images, achieving 96.8% precision through data augmentation and TensorFlow datasets.
- Integrated TensorFlow Lite with a React Native mobile app and FastAPI backend, enabling farmers to capture leaf images and receive real-time disease predictions; deployed the system on Google Cloud Functions for scalable access.